10:32 am

1-Number of Zeros ...
118.185.187.137

**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int zero(int a[], int l, int h) {
    if (l > h) return 0;
    if (a[l] == 0) return h - l + 1;

    int m = (l + h) / 2;

    if (a[m] == 1)
        return zero(a, m + 1, h);
    return zero(a, l, m);
}

int main() {
    int n, a[100];
    scanf("%d", &n);

    for(int i=0; i<n; i++)
        scanf("%d", &a[i]);

    printf("%d", zero(a, 0, n-1));
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 5<br>1<br>1<br>1<br>0<br>0 | 2 | 2 | ✔ |
| ✔ | 10<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1 | 0 | 0 | ✔ |
| ✔ | 8<br>0<br>0<br>0<br>0<br>0<br>0<br>0<br>0 | 8 | 8 | ✔ |
| ✔ | 17<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1 | 2 | 2 | ✔ |

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```
    int m = (l + h) / 2;

    if (a[m] == 1)
        return zero(a, m + 1, h);
    return zero(a, l, m);
}

int main() {
    int n, a[100];
    scanf("%d", &n);

    for(int i=0; i<n; i++)
        scanf("%d", &a[i]);

    printf("%d", zero(a, 0, n-1));
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 5<br>1<br>1<br>1<br>0<br>0 | 2 | 2 | ✔ |
| ✔ | 10<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1 | 0 | 0 | ✔ |
| ✔ | 8<br>0<br>0<br>0<br>0<br>0<br>0<br>0<br>0 | 8 | 8 | ✔ |
| ✔ | 17<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>0<br>0 | 2 | 2 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Back to Course

10:32 am

2-Majority Element...
118.185.187.137

| | |
|---|---|
| Marks | 1.00/1.00 |
| Grade | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00 Flag question

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

**Example 1:**

```
Input: nums = [3,2,3]
Output: 3
```

**Example 2:**

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

**Constraints:**

- `n == nums.length`
- `1 <= n <= 5 * 10^4`
- `-2^31 <= nums[i] <= 2^31 - 1`

**For example:**

| Input | Result |
|---|---|
| 3<br>3 2 3 | 3 |
| 7<br>2 2 1 1 1 2 2 | 2 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n, a[50000], count = 0, candidate;

    scanf("%d", &n);

    for(int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    for(int i = 0; i < n; i++) {
        if(count == 0)
            candidate = a[i];

        if(a[i] == candidate)
            count++;
        else
            count--;
    }

    printf("%d", candidate);

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 3<br>3 2 3 | 3 | 3 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

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Data retention summary

10:33 am

3-Finding Floor Val...
118.185.187.137

**Question 1** | Correct | Mark 1.00 out of 1.00 | Flag question

**Problem Statement:**

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

**Input Format**

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

**Output Format**

First Line Contains Integer – Floor value for x

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int floorValue(int a[], int l, int h, int x)
{
    if (l > h)
        return -1;

    int mid = (l + h) / 2;

    if (a[mid] == x)
        return a[mid];

    if (a[mid] < x)
        return (mid == h) ? a[mid] :
               (a[mid + 1] > x ? a[mid] : floorValue(a, mid + 1, h, x));

    return floorValue(a, l, mid - 1, x);
}

int main()
{
    int n, a[100], x;

    scanf("%d", &n);

    for(int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    scanf("%d", &x);

    printf("%d", floorValue(a, 0, n - 1, x));

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 6<br>1<br>2<br>8<br>10<br>12<br>19<br>5 | 2 | 2 | ✔ |
| ✔ | 5<br>10<br>22<br>85<br>108<br>129<br>100 | 85 | 85 | ✔ |
| ✔ | 7<br>3<br>5<br>7<br>9<br>11<br>13<br>15<br>10 | 9 | 9 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

10:33 am

4-Two Elements su...
118.185.187.137

| | |
|---|---|
| **Time taken** | 2 mins 18 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** Correct Mark 1.00 out of 1.00 Flag question

**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

**Input Format**

First Line Contains Integer n - Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x - Sum Value

**Output Format**

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int find(int a[], int l, int r, int x) {
    if(l >= r) return 0;

    if(a[l] + a[r] == x) {
        printf("%d\n%d", a[l], a[r]);
        return 1;
    }

    if(a[l] + a[r] < x)
        return find(a, l + 1, r, x);

    return find(a, l, r - 1, x);
}

int main() {
    int n, a[100], x;

    scanf("%d", &n);
    for(int i=0; i<n; i++) scanf("%d", &a[i]);
    scanf("%d", &x);

    if(!find(a, 0, n-1, x))
        printf("No");

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 4<br>2<br>4<br>8<br>10<br>14 | 4<br>10 | 4<br>10 | ✔ |
| ✔ | 5<br>2<br>4<br>6<br>8<br>10<br>100 | No | No | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

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Data retention summary

10:33 am

5-Implementation ...
118.185.187.137

| | |
|---|---|
| **Started on** | Friday, 11 September 2026, 8:10 PM |
| **State** | Finished |
| **Completed on** | Friday, 11 September 2026, 8:13 PM |
| **Time taken** | 2 mins 59 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct | Mark 1.00 out of 1.00 | Flag question

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

**For example:**

| Input | Result |
|---|---|
| 5<br>67 34 12 98 78 | 12 34 67 78 98 |

**Answer:**

```c
#include <stdio.h>

void q(int a[],int l,int h){
    if(l>=h) return;
    int i=l,j=h,p=a[l],t;
    while(i<j){
        while(a[i]<=p && i<h)i++;
        while(a[j]>p)j--;
        if(i<j){t=a[i];a[i]=a[j];a[j]=t;}
    }
    a[l]=a[j]; a[j]=p;
    q(a,l,j-1); q(a,j+1,h);
}

int main(){
    int n,a[100];
    scanf("%d",&n);
    for(int i=0;i<n;i++)scanf("%d",&a[i]);
    q(a,0,n-1);
    for(int i=0;i<n;i++)printf("%d ",a[i]);
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 5<br>67 34 12 98 78 | 12 34 67 78 98 | 12 34 67 78 98 | ✔ |
| ✔ | 10<br>1 56 78 90 32 56 11 10 90 114 | 1 10 11 32 56 56 78 90 90 114 | 1 10 11 32 56 56 78 90 90 114 | ✔ |
| ✔ | 12<br>9 8 7 6 5 4 3 2 1 10 11 90 | 1 2 3 4 5 6 7 8 9 10 11 90 | 1 2 3 4 5 6 7 8 9 10 11 90 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

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Data retention summary